F79MA Statistical Models A 2026-27
Assessed Project 1

Project description

It is known that many major geological events occur partly in response to the movement of tectonic plates which cover the Earth's surface. The boundaries of these tectonic plates are of particular interest and are referred to as fault lines. In earthquake modelling, it is interesting to consider whether there is a connection between tremors on either side of a fault line.

Consider two zones, A and B, consistent of regions either side of a fault line. For any day, let Y^A denote the number of tremors in zone A and Y^B denote the number of tremors in zone B. For each day, we will model the pair (Y^A, Y^B) by

$$Y^A = X_0 + X_1$$

and

$$Y^B = X_0 + X_2,$$

where X_0, X_1 and X_2 are independent random variables following Poisson distributions with rates λ_0, λ_1 and λ_2 respectively. We will assume that the distribution of (Y^A, Y^B) is the same for each day and all days are independent of previous days.

Your task is to write a report considering estimating the rate λ_0 for λ_1 and λ_2 known. This requires both theoretical analysis and simulation in R.

To support this, you are provided with the R script `F79MA_Assignment1.R` (available on Canvas). In this script, you must set the argument of `set.seed` equal to the last four digits of your Heriot-Watt student ID number. For example, if your student ID is `H12345678`, then your seed should be `5678`. **Failure to do so may result in 0 marks being awarded.**

This R script will then define a function `simulate_YA_YB` based on your Student ID for simulating an independent and identically distributed (i.i.d.) random sample of size n of values $\mathbf{Y}^A$ and $\mathbf{Y}^B$ assuming $\lambda_0 = 2$, $\lambda_1 = 3$ and $\lambda_2 = 4$. Importantly, note that each pair (Y_i^A, Y_i^B) will not be mutually independent.

Do not change any of the code above the line reading 'Please insert your R code after this line' **except** for the seed. **Failure to do so may result in 0 marks being awarded.**

You must submit, using the links on Canvas: (a) your written report, as a pdf file or as a Word document; and (b) an R script containing code that calculates required quantities and produces any plots you include. Before submitting, you must complete the Standard Declaration of Student Authorship quiz on Canvas.

Tasks

1. Explain carefully what assumptions have been made in modelling the random variables Y^A and Y^B for a day. [2 marks]

2. State and explain the distributions of Y^A and Y^B, and calculate the covariance of Y^A and Y^B. [3 marks]

3. Assume that λ_1 and λ_2 are known. Derive the likelihood $L(\lambda_0; (\mathbf{Y}^A, \mathbf{Y}^B))$, log-likelihood $\ell(\lambda_0; (\mathbf{Y}^A, \mathbf{Y}^B))$ and score function $U(\lambda_0; (\mathbf{Y}^A, \mathbf{Y}^B)) = \frac{\partial \ell}{\partial \lambda_0}$ for a random sample $(\mathbf{Y}^A, \mathbf{Y}^B)$ over n days where

$$\mathbf{Y}^A = (Y_1^A, Y_2^A, \ldots, Y_n^A)$$

and

$$\mathbf{Y}^B = (Y_1^B, Y_2^B, \ldots, Y_n^B).$$

Hint: You should first determine the joint distribution of (Y_i^A, Y_i^B) for each i as the number of tremors across the two zones are not independent. [4 marks]

4. Using R, for a **single** sample of size $n = 30$ of $(\mathbf{Y}^A, \mathbf{Y}^B)$ simulated using the function `simulate_YA_YB` for your seed, plot the observed score function and, using this or otherwise, find the maximum likelihood estimate (MLE) for this sample. Store the value of the MLE as `Quantity1`. [2 marks]

5. Given a random sample $(\mathbf{Y}^A, \mathbf{Y}^B)$ of size n, we may define an alternative estimator T for λ_0 by

$$T(\mathbf{Y}^A, \mathbf{Y}^B) = \frac{\overline{\mathbf{Y}^A} + \overline{\mathbf{Y}^B} - \lambda_1 - \lambda_2}{2},$$

where

$$\overline{\mathbf{Y}^A} = \frac{1}{n}\sum_{i=1}^{n} Y_i^A$$

denotes the sample mean of $\mathbf{Y}^A$, and $\overline{\mathbf{Y}^B}$ is defined similarly.

Derive the bias and variance of the estimator T, and comment. [3 marks]

6. Using R, for a **single** sample of size $n = 30$ of $(\mathbf{Y}^A, \mathbf{Y}^B)$ simulated using the function `simulate_YA_YB` for your seed, calculate T and store the value as `Quantity2`. Comment on how it compares to value of the MLE you have calculated in Task 4 above. [2 marks]

7. Using R, through repeated sampling of samples of size $n = 30$, plot a histogram of the observed values of T and comment on its shape and main characteristics. [3 marks]

   Note: You will need to think carefully how to make use of the function `simulate_YA_YB` to obtain multiple independent samples.

8. Using R, for a **single** sample of size $n = 30$ of $(\mathbf{Y}^A, \mathbf{Y}^B)$ simulated using the function `simulate_YA_YB` for your seed, provide an approximate 95% Confidence Interval for T by making use of your estimate `Quantity2` (i.e. not using the value $\lambda_0 = 2$ used in the simulation) and the distribution obtained via Task 7. Store the values of the lower limit as `Quantity3` and the upper limit as `Quantity4`. [3 marks]

9. Using R, through repeated sampling of samples of size $n = 30$, estimate the Fisher information and hence the Cramer-Rao Lower Bound (CRLB) for an unbiased estimator for λ_0 for a random sample of size n = 30 using $\lambda_0 = 2$. Save the estimated Fisher information $I(2)$ as `Quantity5` and the CRLB as `Quantity6`. Comment on your results. [2 marks]

10. Using R, plot the variance of T against the CRLB for n varying from 10 to 50, and comment. [2 marks]

The Report

Your findings should be presented in the form of a report for your line manager, who is also part of the statistical modelling team but has not worked directly on this problem.

Your report should:

- have a clear and logical structure; include an introduction and clearly stated conclusions that can be understood by any numerate scientist;
- include detail of your mathematical calculations so that your results could be reproduced by another statistician;
- include clearly labelled and correctly referenced tables and diagrams, as appropriate;
- include citation and referencing for any material (books, papers, websites etc) used. When possible, use reliable sources, produced by respected and well-known authors, published by recognised publishers and associated with well established government, academic, or educational institutions. Note that some webpages, YouTube videos, blog posts, and Wikipedia pages might include errors.

Your report should be produced using a word-processing program, such as Microsoft Word, or with LaTeX, and uploaded as a single file in PDF format. All equations should be produced digitally (typeset) by using an equation editor (we recommend to avoid scanned or handwritten equations). You may use the Microsoft Word template provided, but this is not compulsory.

The report has a **maximum page limit of ten (10) pages** (equivalent to 5 double-sided A4 sheets, 11-point font).

A total of **4 Marks** is available for these aspects of your report. This will be marked according to the rubric given in the Appendix.

[Total: 30 Marks]

Notes

- **Errors in code:** Six marks are for correctly carrying out calculations in R (i.e. calculating the correct values for the Quantities), and this will be assessed by running your R code and comparing the results with the verified correct output. You must therefore ensure that you have defined variables as above (i.e. **exactly the same naming convention with capital Q and no underscore between Quantity and the number**) and ensure your code runs without generating errors. **Failure to do so may result in 0 marks being awarded**. You can check your code runs correctly by running it through the file `TestSubmissionAssignment1.R`.

- **Use of libraries and good practice with code:** Please do not give references to directories on your own computer (e.g. specifying a directory with setwd) or make use of any R library aside from the 'ggplot2' library (you do not need to use this library). Your code should be commented appropriately to ease with checking.

- **Questions for the Lecturers**: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly. That way all students will see the same responses. The discussion board will be closed at 9:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time on Friday 30th October 2026.

- **Use of generative artificial intelligence (Gen AI)**: You may make use of Gen AI such as large language models for thinking through your strategy for solving the tasks, but the report must be written in your own words and all code must be your own. You should note your usage of Gen AI within your report, e.g. in an appendix or with the references.

  Note that Gen AI is not required for this assignment, nor are other external references such as books or websites, but any references used should be cited appropriately.

- **Collaboration and Plagiarism**: Students are allowed to discuss the methods used with other students, but your submitted project must be **all your own work**. Particularly please note the following:
  - Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
  - Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course or worse.
  - Students must never give hard or soft copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
  - Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

  More information on plagiarism can be found here
- **Submission**: The report and the code should be submitted through Canvas using the appropriate links. It should not be submitted by email or handed in as a paper copy. You should also complete the Student Declaration of Authorship quiz.
- **Deadline**: The deadline for submission is **9:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time** on **Monday 2nd November 2026** (Week 8); projects may be submitted early. Late projects will be marked in line with the University Coursework Policy, i.e. a standard deduction of 30% if submitted within 7 days and 0 marks awarded if submitted later. You can, however, submit a mitigating circumstances applications, but these may only be applied at the end of the semester. No individual extensions are allowed.
- **Feedback**: It is expected that feedback on this assignment will be available by Monday 23rd November 2026 (Week 11).
- **Assessment**: This coursework will contribute **20**% to the final mark for the course.

Appendix: Rubric for marking of the report

The five marks available for the exposition of your report will be awarded according to the scale below:

Between 0 and 1 marks will be awarded for	• Lack of clear and logical structure • Conclusions missing or not suitable for a non-statistician • Statistical calculations and methodology not clearly set out for the reader • Tables and figures unclear, badly labelled or not correctly referred to • Any sources or AI used not clearly referenced
Between 2 and 3 marks will be awarded for	• Clear and logical structure including Introduction and Conclusion • Conclusions generally suitable for a non-statistician • Statistical calculations and methodology generally set out clearly for the reader • Tables and figures often clear and correctly referred to • Any sources or AI used clearly referenced
4 marks will be awarded for	• Clear and logical structure including Introduction and Conclusion • Conclusions suitable for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures clear, correctly referred to and easy to interpret • Any sources or AI used clearly and correctly referenced